

Open Data Fusion for Approximate True Cost Accounting using Knowledge Graphs

Contents

1. Abstract	3
2. Introduction	3
3. Related Work	3
4. Methodology	3
4.1. Graph Construction	3
4.1.1. Source: Wikirate	3
4.1.2. Source: Openproductfacts	3
5. Graph Utilization	3
6. Evaluation	3
7. Discussion	3
8. Appendix	3
8.1. Graph Schema	3
Bibliography	4

1. Abstract

2. Introduction

3. Related Work

4. Methodology

At the core of this work lies the knowledge graph, which the platform surrounding it attempts to make easily more accessible. This section is therefore divided into graph construction (including maintenance) and graph utilization.

4.1. Graph Construction

The data ingestion phase is designed to be flexible and extensible to serve different use cases. Inclusion of sources can be toggled in the interface. With the currently available sources graph construction and indexing takes around a minute on consumer hardware. This is subject to change with the inclusion of more or remote sources. After adding the schema (see Appendix 8.1), each source is parsed into triples and stored using the graph database oxigraph (CITATION)

The available sources are described in the following.

4.1.1. Source: Wikirate

Wikirate [1] is an open data platform, which crowdsources and hosts a variety of company data relating to Environmental Social and Governance (ESG) issues. Every data point regardless of type (Company, Metric, Project, Tag, Image) lives in a shared namespace as so called cards.

4.1.2. Source: Openproductfacts

5. Graph Utilization

6. Evaluation

7. Discussion

8. Appendix

8.1. Graph Schema

```
@prefix : <http://hiddencostreport.org/schema#> .
@prefix rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#> .
@prefix rdfs: <http://www.w3.org/2000/01/rdf-schema#> .
@prefix owl: <http://www.w3.org/2002/07/owl#> .
@prefix dc: <http://purl.org/dc/elements/1.1/> .
```

```
@prefix wr: <https://wikirate.org> .
```

```
:Company a rdfs:Class .
:Product a rdfs:Class .
:Article a rdfs:Class .
:Metric a rdfs:Class .
:Observation a rdfs:Class .
:Dataset a rdfs:Class .
:Datapoint a rdfs:Class .
```

```
:Product a rdfs:Class .

:Name a rdf:property ;
  rdfs:range rdfs:literal .

:OpenCorporatesID a rdf:property ;
  rdfs:domain :Company ;
  rdfs:range rdfs:literal .

:MetricTitle a rdf:property ;
  rdfs:domain :Metric ;
  rdfs:range rdfs:literal .

:MetricDesigner a rdf:property ;
  rdfs:domain :Metric ;
  rdfs:range rdfs:literal .

:Questions a rdf:property ;
  rdfs:domain :Metric ;
  rdfs:range rdfs:literal .

:ValueType a rdf:property ;
  rdfs:domain :Metric ;
  rdfs:range rdfs:literal .

:Unit a rdf:property ;
  rdfs:domain :Metric ;
  rdfs:range rdfs:literal .

:hasMetric a rdf:property;
  rdfs:domain :Company ;
  rdfs:range :Observation .

:Brand a rdf:property ;
  rdfs:domain :Product .

:Ingredient a rdf:property ;
  rdfs:domain :Product .

:ManufacturedIn a rdf:property ;
  rdfs:domain :Product .
```

Bibliography

[1] “Wikirate.”